

LUKA, Czech Republic

WMO: 117100

Lat: 49.6522N

Lon: 16.9533E

Elev: 516

StdP: 95.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.5	-16.6	0.9	-11.5	-13.9	1.2	-9.8	15.0	2.7	13.3	2.2	3.3	40	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.0	18.7	27.0	18.2	25.3	17.5	19.9	26.1	19.0	25.1	18.2	23.9	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.5	22.6	16.8	12.8	21.5	16.1	12.1	20.5	59.2	26.2	56.1	25.2	53.3	23.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	9.0	-15.1	31.5	3.2	2.2	-17.4	33.1	-19.3	34.4	-21.1	35.7	-23.5	37.3		
(5)			-15.4	21.6	3.2	1.0	-17.7	22.3	-19.6	23.0	-21.4	23.5	-23.8	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-2.3	-0.6	3.1	8.7	13.0	16.4	18.5	18.4	13.5	8.4	3.6	-1.2		
	DBStd	8.43	4.54	4.68	4.38	4.50	3.78	3.72	3.54	3.41	3.58	4.02	4.12	4.22		
	HDD10.0	1617	382	298	217	75	14	1	0	0	9	80	193	348		
	HDD18.3	3770	640	531	473	289	168	81	42	42	149	308	441	606		
	CDD10.0	1013	0	0	2	37	108	194	265	261	114	31	3	0		
	CDD18.3	124	0	0	0	0	4	23	49	44	4	0	0	0		
	CDH23.3	877	0	0	0	2	23	148	371	312	22	0	0	0		
	CDH26.7	199	0	0	0	0	1	23	98	76	2	0	0	0		
(14) Wind	WSAvg	4.3	4.4	4.7	4.8	4.4	4.3	4.0	4.1	3.7	4.1	4.0	4.5	4.4		
(15) Precipitation	PrecAvg	599	33	28	40	39	64	72	83	73	54	39	40	34		
	PrecMax	808	57	79	98	82	156	125	270	153	124	98	87	62		
	PrecMin	451	7	4	6	5	23	11	4	29	12	4	0	13		
	PrecStd	75	15	17	23	20	29	28	50	36	31	22	18	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	10.9	16.1	22.6	25.7	29.1	31.7	31.4	26.1	19.8	14.4	9.4		
		MCWB	7.2	7.0	9.3	13.9	16.8	19.4	18.5	18.4	16.4	14.8	11.2	7.6		
	2%	DB	7.3	8.7	13.5	20.0	23.6	26.9	29.4	28.7	23.2	17.5	12.2	7.2		
		MCWB	5.5	6.2	8.1	12.4	16.0	18.6	18.8	18.3	16.0	13.5	10.1	5.8		
	5%	DB	5.1	7.0	11.3	17.8	21.6	25.0	27.2	26.5	21.1	15.7	10.6	5.6		
		MCWB	3.8	4.8	7.0	11.2	15.0	17.7	18.1	17.9	15.1	12.6	9.0	4.5		
	10%	DB	3.4	5.2	9.2	15.7	19.6	23.1	25.3	24.7	19.3	14.0	9.1	4.1		
		MCWB	2.4	3.6	6.0	10.2	13.9	16.6	17.5	17.4	14.3	11.7	7.8	3.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	7.9	10.1	14.6	18.1	21.0	21.1	20.7	18.2	16.0	11.7	7.9		
		MCDB	9.1	10.1	14.4	21.0	23.9	27.4	27.6	27.0	23.5	18.2	13.2	8.9		
	2%	WB	5.7	6.4	8.7	13.1	16.7	19.6	19.9	19.6	16.8	14.1	10.4	6.0		
		MCDB	6.9	8.3	12.4	18.9	22.2	25.0	26.5	25.8	21.3	16.4	12.0	7.0		
	5%	WB	3.9	5.2	7.6	11.8	15.6	18.4	19.0	18.7	15.8	13.0	9.1	4.7		
		MCDB	5.0	6.8	10.6	16.7	20.5	23.6	25.4	24.8	20.0	15.1	10.4	5.5		
	10%	WB	2.5	3.7	6.4	10.6	14.5	17.3	18.1	17.8	14.9	12.0	7.9	3.4		
		MCDB	3.4	5.0	8.8	14.9	18.7	22.1	23.9	23.6	18.6	13.8	9.0	4.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.1	6.6	8.5	8.7	9.0	9.4	9.5	7.6	6.0	4.3	3.8		
		MCDBR	5.6	6.7	9.7	10.8	10.9	11.2	12.1	12.0	10.3	8.0	5.7	5.0		
	5% WB	MCWBR	4.5	4.7	5.8	5.7	5.7	5.3	4.7	4.7	4.8	4.7	4.1	4.1		
		MCWBR	5.3	6.1	8.8	10.2	10.1	10.5	11.2	10.8	9.3	7.4	5.2	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.319	0.355	0.400	0.399	0.399	0.412	0.407	0.378	0.353	0.330	0.295		
		taud	2.346	2.332	2.298	2.229	2.266	2.293	2.261	2.290	2.347	2.393	2.401	2.394		
	Ebn at Noon		749	820	847	843	860	860	841	828	814	770	700	700		
		Edh at Noon	67	87	107	128	128	126	128	119	101	82	65	56		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.64	2.74	4.22	5.04	5.41	5.16	4.63	3.27	1.89	0.92	0.65		
	RadStd		0.15	0.22	0.47	0.50	0.61	0.48	0.55	0.43	0.47	0.28	0.11	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		+0.53	N/A	N/A	+0.79	+0.41	+0.36	-122	-160	+74	+30

Nomenclature: See separate page